Mini-Lecture: The Gaussian AWGN Case Study

A Step-by-Step Walkthrough of Section 5 of
"Generative Modeling from Black-box Corruptions
via Self-Consistent Stochastic Interpolants"

Prepared as a pedagogical companion

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1 What This Section Is About

The paper introduces a method called **Self-Consistent Stochastic Interpolants (SCSI)** for learning a generative model of clean data π when you only have access to corrupted observations μ and a black-box forward corruption model $\mathcal{F}$.

Section 5 takes the *simplest possible case* — a Gaussian prior corrupted by additive white Gaussian noise (AWGN) — and works out *everything* in closed form. This lets us:

(i) Verify that the iterative scheme actually converges.

(ii) Compute the exact convergence rate.

(iii) Compare SCSI (with ODE transport) to the classical EM algorithm.

(iv) Check the theoretical assumptions from Section 4.

2 Setup

2.1 The True Data Distribution

We assume the clean data follows a Gaussian distribution:

$$\pi = \mathcal{N}(b, \Sigma), \qquad \Sigma \succ 0 \text{ (positive definite)}.$$

2.2 The Forward (Corruption) Channel

The AWGN channel adds independent Gaussian noise:

$$y = \mathcal{F}(x) = x + w, \qquad w \sim \mathcal{N}(0, I).$$

Intuition

We've *normalized* the noise variance to I without loss of generality. If the original channel were $y = x + \sigma\xi$ with $\xi \sim \mathcal{N}(0, I)$, we could rescale $x \to x/\sigma$ to get unit noise. The "signal-to-noise ratio" (SNR) is then encoded in Σ — eigenvalues of Σ that are large relative to 1 correspond to high SNR directions, and small eigenvalues correspond to low SNR.

2.3 The Observation Distribution

Since $y = x + w$ with $x \sim \mathcal{N}(b, \Sigma)$ and $w \sim \mathcal{N}(0, I)$ independent:

$$\mu = \mathrm{Law}(y) = \mathcal{N}(b, \Sigma + I).$$

This is just the convolution $\mu = \pi \star \gamma_1$, where $\gamma_1 = \mathcal{N}(0, I)$.

3 The Stochastic Interpolant for This Problem

3.1 Choice of Interpolant Schedule

The paper uses the specific interpolant:

$$I_t = (1 - \sqrt{t})\, x + \sqrt{t}\, \mathcal{F}(x) = (1 - \sqrt{t})\, x + \sqrt{t}\, (x + w) = x + \sqrt{t}\, w.$$

Simplifying the interpolant

Let's verify this step by step. We have $\alpha_t = 1 - \sqrt{t}$ and $\beta_t = \sqrt{t}$, and we set $\gamma_t = 0$ (no extra noise term). Then:

$$\begin{aligned} I_t &= \alpha_t\, x + \beta_t\, \mathcal{F}(x) \\ &= (1 - \sqrt{t})\, x + \sqrt{t}\,(x + w) \\ &= x - \sqrt{t}\, x + \sqrt{t}\, x + \sqrt{t}\, w \\ &= x + \sqrt{t}\, w. \end{aligned} \tag{1}$$

This is a beautiful simplification: the interpolant is just the clean signal x plus a *scaled* version of the noise w.

3.2 The Law of I_t

Key Idea

For a *general* Gaussian measure $\tilde{\pi} = \mathcal{N}(\tilde{b}, \tilde{\Sigma})$ replacing π, the law of I_t is:

$$\tilde{\pi}_t = \mathrm{Law}(I_t) = \mathcal{N}(\tilde{b},\, \tilde{\Sigma} + t\, I).$$

Deriving the law of I_t

If $x \sim \mathcal{N}(\tilde{b}, \tilde{\Sigma})$ and $w \sim \mathcal{N}(0, I)$ are independent, then $I_t = x + \sqrt{t}\, w$ is a sum of independent Gaussians:

$$\begin{aligned} \mathbb{E}[I_t] &= \mathbb{E}[x] + \sqrt{t}\, \mathbb{E}[w] = \tilde{b} + 0 = \tilde{b}, \\ \mathrm{Cov}(I_t) &= \mathrm{Cov}(x) + t\, \mathrm{Cov}(w) = \tilde{\Sigma} + t\, I. \end{aligned}$$

So indeed $I_t \sim \mathcal{N}(\tilde{b},\, \tilde{\Sigma} + tI)$.

Boundary conditions:

- At $t = 0$: $I_0 = x \sim \mathcal{N}(\tilde{b}, \tilde{\Sigma})$ ✓ (the "clean" distribution).
- At $t = 1$: $I_1 = x + w \sim \mathcal{N}(\tilde{b}, \tilde{\Sigma} + I)$ ✓ (the "corrupted" distribution $\mathcal{K}\tilde{\pi}$).

3.3 Connection to the Heat Equation

The evolution $\tilde{\pi}_t = \mathcal{N}(\tilde{b}, \tilde{\Sigma} + tI)$ corresponds exactly to the **heat equation**:

$$\partial_t \tilde{\pi}_t = \tfrac{1}{2} \Delta \tilde{\pi}_t, \qquad \tilde{\pi}_0 = \tilde{\pi}.$$

Intuition

Running the heat equation for time t on a Gaussian $\mathcal{N}(\tilde{b}, \tilde{\Sigma})$ simply adds tI to the covariance — this is exactly what Gaussian convolution does. The forward corruption process "is" the heat equation from $t = 0$ to $t = 1$.

4 The Score Function

For a Gaussian $\tilde{\pi}_t = \mathcal{N}(\tilde{b}, \tilde{\Sigma} + tI)$, the score (gradient of the log-density) is:

$$\nabla \log \tilde{\pi}_t(x) = -(\tilde{\Sigma} + tI)^{-1}(x - \tilde{b}).$$

Deriving the Gaussian score

The density of $\mathcal{N}(\tilde{b}, C)$ with $C = \tilde{\Sigma} + tI$ is:

$$\tilde{\pi}_t(x) = \frac{1}{(2\pi)^{d/2} \det(C)^{1/2}} \exp\Big(-\tfrac{1}{2}(x - \tilde{b})^\top C^{-1}(x - \tilde{b})\Big).$$

Taking the log:

$$\log \tilde{\pi}_t(x) = \text{const} - \tfrac{1}{2}(x - \tilde{b})^\top C^{-1}(x - \tilde{b}).$$

Taking the gradient with respect to x:

$$\nabla_x \log \tilde{\pi}_t(x) = -C^{-1}(x - \tilde{b}) = -(\tilde{\Sigma} + tI)^{-1}(x - \tilde{b}).$$

This is an **affine** function of x, which will be crucial for keeping everything in closed form.

5 The Reverse Transport SDE

5.1 General Form

The paper uses a reverse-time SDE to transport observations (at $t = 1$) back to clean data (at $t = 0$). In the notation of the paper, switching to the reverse time variable $\tilde{t}$ (where $\tilde{t} = 0$ is the observation end and $\tilde{t} = 1$ is the data end):

$$dY_{\tilde{t}} = \frac{1 + \epsilon}{2} \nabla \log \tilde{\pi}_{1-\tilde{t}}(Y_{\tilde{t}})\, d\tilde{t} + \sqrt{\epsilon}\, dW_{\tilde{t}}.$$

Where does this SDE come from?

The general reverse-time SDE from Section 2.1 (Eq. 2 in the paper) has drift $b + \epsilon\gamma_t^{-1} g$, where b is the velocity field and g is the denoiser. For the heat equation channel, the connection between score and these fields gives a total drift of $\frac{1+\epsilon}{2}\nabla \log \tilde{\pi}_{1-\tilde{t}}$.

The key point: the reverse process is an Ornstein–Uhlenbeck (OU) process because $\nabla \log \tilde{\pi}_{1-\tilde{t}}$ is *affine* in x (see previous section). OU processes can be solved exactly.

Special case $\epsilon = 0$ (ODE):

$$dY_{\tilde{t}} = \tfrac{1}{2}\nabla \log \tilde{\pi}_{1-\tilde{t}}(Y_{\tilde{t}})\, d\tilde{t}.$$

This is a deterministic ODE — no randomness, just follow the score.

5.2 Substituting the Gaussian Score

Substituting the score $\nabla \log \tilde{\pi}_{1-\tilde{t}}(x) = -(\tilde{\Sigma} + (1 - \tilde{t})I)^{-1}(x - \tilde{b})$:

$$dY_{\tilde{t}} = -\frac{1 + \epsilon}{2}(\tilde{\Sigma} + (1 - \tilde{t})I)^{-1}(Y_{\tilde{t}} - \tilde{b})\, d\tilde{t} + \sqrt{\epsilon}\, dW_{\tilde{t}}.$$

This is a **time-varying linear SDE** (Ornstein–Uhlenbeck type) and can be solved in closed form.

6 Solving the Reverse SDE: The Solution Map

6.1 The Fundamental Solution Map $\Phi(t, s)$

The paper defines the solution (or "propagator") map:

$$\boxed{\Phi(t,s) = \big(\tilde{\Sigma} + (1-t)I\big)^{(1+\epsilon)/2}\big(\tilde{\Sigma} + (1-s)I\big)^{-(1+\epsilon)/2}}$$

which maps the state from time s to time t.

Verifying that Φ solves the homogeneous equation

Consider the homogeneous (noise-free) part of the SDE. We need to solve:

$$\frac{d}{d\tilde{t}}\,\Phi(\tilde{t}, s) = -\frac{1+\epsilon}{2}(\tilde{\Sigma} + (1-\tilde{t})I)^{-1}\,\Phi(\tilde{t}, s), \qquad \Phi(s, s) = I.$$

Since all the matrices $\tilde{\Sigma} + (1-\tilde{t})I$ commute with each other for different values of $\tilde{t}$ (they are all functions of the *same* matrix $\tilde{\Sigma}$), we can use the matrix exponential approach.

Key trick: Define $A(\tilde{t}) = \tilde{\Sigma} + (1-\tilde{t})I$. Note that:

$$\frac{d}{d\tilde{t}}\,A(\tilde{t})^{(1+\epsilon)/2} = \frac{1+\epsilon}{2}\,A(\tilde{t})^{(1+\epsilon)/2-1}\cdot(+I) = \frac{1+\epsilon}{2}\,A(\tilde{t})^{(1+\epsilon)/2}\,A(\tilde{t})^{-1},$$

where we used the chain rule and $\frac{d}{d\tilde{t}}A(\tilde{t}) = -(-I) = +I$...

Wait, let's be more careful. We have $A(\tilde{t}) = \tilde{\Sigma} + (1-\tilde{t})I$, so:

$$\frac{dA}{d\tilde{t}} = -I.$$

Now, for a matrix function $f(A) = A^p$ where A depends on a scalar parameter and all values of A commute:

$$\frac{d}{d\tilde{t}}A(\tilde{t})^p = p\,A(\tilde{t})^{p-1}\frac{dA}{d\tilde{t}} = p\,A(\tilde{t})^{p-1}(-I) = -p\,A(\tilde{t})^{p-1}.$$

So with $p = (1+\epsilon)/2$:

$$\frac{d}{d\tilde{t}}A(\tilde{t})^{(1+\epsilon)/2} = -\frac{1+\epsilon}{2}\,A(\tilde{t})^{(1+\epsilon)/2-1}.$$

Now compute:

$$\begin{aligned}
\frac{d}{d\tilde{t}}\Phi(\tilde{t}, s) &= \frac{d}{d\tilde{t}}\Big[A(\tilde{t})^{(1+\epsilon)/2}\Big]\cdot A(s)^{-(1+\epsilon)/2} \\
&= -\frac{1+\epsilon}{2}\,A(\tilde{t})^{(1+\epsilon)/2-1}\cdot A(s)^{-(1+\epsilon)/2} \\
&= -\frac{1+\epsilon}{2}\,A(\tilde{t})^{-1}\cdot\underbrace{A(\tilde{t})^{(1+\epsilon)/2}\cdot A(s)^{-(1+\epsilon)/2}}_{\Phi(\tilde{t},s)} \\
&= -\frac{1+\epsilon}{2}\,A(\tilde{t})^{-1}\,\Phi(\tilde{t}, s).
\end{aligned}$$

This is exactly the ODE $\dot{\Phi} = -\frac{1+\epsilon}{2}(\tilde{\Sigma} + (1-\tilde{t})I)^{-1}\Phi$. Also $\Phi(s, s) = A(s)^{(1+\epsilon)/2}A(s)^{-(1+\epsilon)/2} = I$. ✓

6.2 The Full Solution via Variation of Constants

Using the solution map, the full SDE solution from $\tilde{t} = 0$ (observations) to $\tilde{t} = 1$ (data) is:

$$Y_1 = \tilde{b} + \Phi(1,0)(Y_0 - \tilde{b}) + \sqrt{\epsilon} \int_0^1 \Phi(1,s)\, dW_s. \tag{2}$$

Deriving Eq. (2) by variation of constants

Write $Z_{\tilde{t}} = Y_{\tilde{t}} - \tilde{b}$ (center the process). Then Z satisfies:

$$dZ_{\tilde{t}} = -\frac{1+\epsilon}{2} A(\tilde{t})^{-1} Z_{\tilde{t}}\, d\tilde{t} + \sqrt{\epsilon}\, dW_{\tilde{t}}.$$

The variation-of-constants formula for a linear SDE $dZ = F(\tilde{t}) Z\, d\tilde{t} + \sigma\, dW$ gives:

$$Z_{\tilde{t}} = \Phi(\tilde{t},0)\, Z_0 + \sqrt{\epsilon} \int_0^{\tilde{t}} \Phi(\tilde{t},s)\, dW_s.$$

At $\tilde{t} = 1$:

$$Z_1 = \Phi(1,0)\, Z_0 + \sqrt{\epsilon} \int_0^1 \Phi(1,s)\, dW_s.$$

Substituting back $Y = Z + \tilde{b}$:

$$Y_1 = \tilde{b} + \Phi(1,0)(Y_0 - \tilde{b}) + \sqrt{\epsilon} \int_0^1 \Phi(1,s)\, dW_s.$$

Intuition

Formula (2) has three parts:

1. $\tilde{b}$: the mean of the current prior guess.
2. $\Phi(1,0)(Y_0 - \tilde{b})$: a "shrunken" version of how far the observation deviates from the mean. The map $\Phi(1,0)$ is a contraction that pulls Y_0 toward $\tilde{b}$.
3. The stochastic integral: noise injected during the reverse diffusion (absent when $\epsilon = 0$).

7 The Iterative Update Rules

7.1 How the Iteration Works

Goal 1. We start with an initial guess $\pi_0 = \mathcal{N}(b_0, \Sigma_0)$ and want to iteratively converge to $\pi = \mathcal{N}(b, \Sigma)$.

At iteration k, our current guess is $\pi_k = \mathcal{N}(b_k, \Sigma_k)$.

One iteration of SCSI:

1. Take observation $Y_0 \sim \mu = \mathcal{N}(b, \Sigma + I)$ (these are the real corrupted data).
2. Run the reverse SDE (2) with $\tilde{b} = b_k$, $\tilde{\Sigma} = \Sigma_k$ to get $Y_1 = \pi_{k+1}$.

7.2 Computing π_{k+1}

Since $Y_0 \sim \mathcal{N}(b, \Sigma + I)$ and the transformation (2) is affine, Y_1 is also Gaussian, so $\pi_{k+1} = \mathcal{N}(b_{k+1}, \Sigma_{k+1})$.

7.2.1 The Matrix B_k

Define:

$$\boxed{B_k = \left(\Sigma_k(\Sigma_k + I)^{-1}\right)^{(1+\epsilon)/2}}$$

Where does B_k come from?

Recall $\Phi(1,0) = A(1)^{(1+\epsilon)/2} A(0)^{-(1+\epsilon)/2}$ with $A(\tilde{t}) = \Sigma_k + (1-\tilde{t})I$.

- $A(0) = \Sigma_k + I$
- $A(1) = \Sigma_k + 0 \cdot I = \Sigma_k$

Therefore:

$$\begin{aligned}\Phi(1,0) &= \Sigma_k^{(1+\epsilon)/2} (\Sigma_k + I)^{-(1+\epsilon)/2} \\ &= \left(\Sigma_k(\Sigma_k + I)^{-1}\right)^{(1+\epsilon)/2} \\ &= B_k.\end{aligned}$$

The last equality uses the fact that Σ_k and $\Sigma_k + I$ commute (they are simultaneously diagonalizable), so we can combine the exponents.

Observation: Since $0 \prec \Sigma_k(\Sigma_k + I)^{-1} \prec I$ (all eigenvalues are between 0 and 1), we have $0 \prec B_k \prec I$ for any $\epsilon \geq 0$.

7.2.2 Mean Update

Deriving b_{k+1}

From (2):

$$\begin{aligned} b_{k+1} &= \mathbb{E}[Y_1] = b_k + B_k(\mathbb{E}[Y_0] - b_k) + 0 \\ &= b_k + B_k(b - b_k). \end{aligned} \tag{3}$$

Here we used:

- $\mathbb{E}[Y_0] = b$ (the true mean of the observations $\mu = \mathcal{N}(b, \Sigma + I)$; note the mean of μ is b, not b_k).
- The stochastic integral has zero mean.

We can rewrite this as:

$$b_{k+1} - b = (I - B_k)(b_k - b),$$

or equivalently, defining $r_k = b - b_k$ (the mean error):

$$\boxed{r_{k+1} = (I - B_k)\, r_k.}$$

Since $0 \prec B_k \prec I$, we have $0 \prec I - B_k \prec I$, which means $\|r_{k+1}\| < \|r_k\|$ — the mean converges!

7.2.3 Covariance Update

Deriving Σ_{k+1}

From (2), $Y_1 - b_{k+1}$ equals:

$$B_k(Y_0 - b) + \sqrt{\epsilon} \int_0^1 \Phi(1, s)\, dW_s.$$

So the covariance is (since Y_0 and W are independent):

$$\begin{aligned} \Sigma_{k+1} &= B_k \operatorname{Cov}(Y_0)\, B_k + \epsilon \int_0^1 \Phi(1,s)\Phi(1,s)^\top\, ds \\ &= B_k\, (\Sigma + I)\, B_k + \epsilon \int_0^1 \Phi(1,s)^2\, ds. \end{aligned} \tag{4}$$

(Note: all matrices commute, so $\Phi(1,s)^\top = \Phi(1,s)$.)

Now, one can show that the stochastic integral term combines with the first term to give the clean formula. Indeed, the paper states the result as:

$$\boxed{\Sigma_{k+1} = \Sigma_k + B_k(\Sigma - \Sigma_k)B_k.}$$

Let's verify this is consistent. Expanding:

$$\begin{aligned} \Sigma_k + B_k(\Sigma - \Sigma_k)B_k &= \Sigma_k + B_k\Sigma B_k - B_k\Sigma_k B_k \\ &= (I - B_k^2)\Sigma_k + B_k\Sigma B_k. \end{aligned}$$

Now, from (4), we need:

$$B_k(\Sigma + I)B_k + (\text{noise term}) = (I - B_k^2)\Sigma_k + B_k\Sigma B_k.$$

The noise term must equal:

$$(I - B_k^2)\Sigma_k + B_k \Sigma B_k - B_k(\Sigma + I)B_k = (I - B_k^2)\Sigma_k - B_k^2.$$

Using $B_k^2 = (\Sigma_k(\Sigma_k + I)^{-1})^{1+\epsilon}$, and recalling that Σ_k and B_k commute, one can verify this identity holds (the Itô integral contribution exactly fills the gap). The full calculation is somewhat involved, so the paper takes this as the result.

Defining the covariance error $R_k = \Sigma - \Sigma_k$:

$$R_{k+1} = \Sigma - \Sigma_{k+1} = R_k - B_k R_k B_k.$$

8 Convergence Analysis

8.1 Contraction of the Covariance Error

We need to show that $\|R_{k+1}\| \le c\,\|R_k\|$ for some $c < 1$.

Bounding $\|R_{k+1}\| = \|R_k - B_k R_k B_k\|$

Claim: For any symmetric matrix X and symmetric B with $0 \prec B \prec I$:

$$\|X - BXB\| \le (1 - \lambda_{\min}(B)^2)\,\|X\|.$$

Proof sketch: Decompose $X = X_+ + X_-$ where $X_+ \succeq 0$ and $X_- \preceq 0$ (positive and negative parts in the spectral decomposition). Then:

$$\begin{aligned} X_+ - BX_+B \succeq 0 \quad &\text{and} \quad X_+ - BX_+B \preceq (1 - \lambda_{\min}(B)^2)\,X_+, \\ X_- - BX_-B \preceq 0 \quad &\text{and} \quad X_- - BX_-B \succeq (1 - \lambda_{\min}(B)^2)\,X_-. \end{aligned}$$

The first line holds because for any v: $v^\top(X_+ - BX_+B)v = v^\top X_+ v - (Bv)^\top X_+(Bv) \le (1 - \lambda_{\min}(B)^2)\,v^\top X_+ v$. Combining gives $\|X - BXB\| \le (1 - \lambda_{\min}(B)^2)\|X\|$.

Applying this: We need a lower bound on $\lambda_{\min}(B_k)$.

8.2 Lower Bound on $\lambda_{\min}(B_k)$

Showing $\lambda_{\min}(\Sigma_k) \ge \eta$ for all k

Define $\eta = \min(\lambda_{\min}(\Sigma), \lambda_{\min}(\Sigma_0))$ and assume $\eta > 0$.

Proof by induction. Suppose $\lambda_{\min}(\Sigma_k) \ge \eta$. From the update:

$$\begin{aligned} \Sigma_{k+1} &= (I - B_k^2)\Sigma_k + B_k \Sigma B_k \\ &\succeq \lambda_{\min}(\Sigma_k)(I - B_k^2) + \lambda_{\min}(\Sigma)\,B_k^2 \\ &\succeq \min\big(\lambda_{\min}(\Sigma_k),\ \lambda_{\min}(\Sigma)\big)\,I \\ &\ge \eta\, I. \end{aligned}$$

The second line uses $\Sigma_k \succeq \lambda_{\min}(\Sigma_k)\,I$ and $\Sigma \succeq \lambda_{\min}(\Sigma)\,I$. The third line uses $(I - B_k^2) + B_k^2 = I$.

Therefore:

$$\lambda_{\min}(B_k) = \left(\frac{\lambda_{\min}(\Sigma_k)}{1+\lambda_{\min}(\Sigma_k)}\right)^{(1+\epsilon)/2} \geq \left(\frac{\eta}{1+\eta}\right)^{(1+\epsilon)/2}.$$

8.3 The Convergence Rate

Putting it all together:

Proposition 1 (Proposition 13 in the paper)**.** *Let* $\eta = \min(\lambda_{\min}(\Sigma), \lambda_{\min}(\Sigma_0)) > 0$. *Then:*

$$\|\Sigma - \Sigma_k\| \leq \left(1 - \eta^{1+\epsilon}(1+\eta)^{-1-\epsilon}\right)^k \|\Sigma - \Sigma_0\|,$$

$$\|b - b_k\|^2 \leq \left(1 - \eta^{1+\epsilon}(1+\eta)^{-1-\epsilon}\right)^k \|b - b_0\|^2.$$

Deriving the rate

The contraction factor per iteration is:

$$c = 1 - \lambda_{\min}(B_k)^2 \leq 1 - \left(\frac{\eta}{1+\eta}\right)^{1+\epsilon}.$$

This is because $\lambda_{\min}(B_k)^2 = \left(\frac{\lambda_{\min}(\Sigma_k)}{1+\lambda_{\min}(\Sigma_k)}\right)^{1+\epsilon} \geq \left(\frac{\eta}{1+\eta}\right)^{1+\epsilon}$.

Writing $\left(\frac{\eta}{1+\eta}\right)^{1+\epsilon} = \eta^{1+\epsilon}(1+\eta)^{-(1+\epsilon)}$, we get the rate stated above.

9 ODE vs. SDE: Why $\epsilon = 0$ Is Optimal

Key Idea

The convergence rate is $\left(1 - \eta^{1+\epsilon}(1+\eta)^{-1-\epsilon}\right)^k$. As a function of ϵ, the factor we subtract is:

$$\left(\frac{\eta}{1+\eta}\right)^{1+\epsilon}.$$

Since $\frac{\eta}{1+\eta} < 1$, increasing ϵ makes this quantity *smaller*, which makes the contraction *worse*. The optimal choice is $\epsilon = 0$ (ODE), giving the contraction factor:

$$1 - \frac{\eta}{1+\eta} = \frac{1}{1+\eta}.$$

Intuition

Why does the ODE beat the SDE? The SDE injects extra noise during the reverse transport. While this can help with robustness in harder problems (e.g., the two-moons synthetic example with high noise), for the Gaussian case the added stochasticity is *pure overhead* that slows convergence.

9.1 Behavior in Different SNR Regimes

- **Low SNR ($\eta \ll 1$):** The signal is weak. The ODE rate is $(1-\eta)^k$ while the EM/SDE($\epsilon = 1$) rate is $(1-\eta^2)^k$. Since $\eta^2 \ll \eta$, ODE converges *much* faster.

  Example: If $\eta = 0.01$ (very noisy), after $k = 100$ iterations:

  - ODE error factor: $(1-0.01)^{100} \approx 0.37$
  - EM error factor: $(1-0.0001)^{100} \approx 0.99$ (barely moved!)

- **High SNR ($\eta \gg 1$):** The contraction is $1/(1+\eta) \ll 1$, so convergence is fast regardless. The corruption is mild, and one iteration nearly recovers π.

10 Comparison with the EM Algorithm

10.1 What Does EM Do?

The EM algorithm from [RALL24] updates a current prior $\tilde{\pi}_k = \mathcal{N}(0, \tilde{\Sigma}_k)$ (focusing on centered Gaussians for simplicity) via:

1. **E-step:** Compute the posterior $p_k(x|y)$. For Gaussian prior + Gaussian noise:

$$p_k(x|y) = \mathcal{N}\big(\tilde{\Sigma}_k(\tilde{\Sigma}_k + I)^{-1}y,\ \ \tilde{\Sigma}_k - \tilde{\Sigma}_k(\tilde{\Sigma}_k + I)^{-1}\tilde{\Sigma}_k\big).$$

2. **M-step:** The new prior is the marginal of the posterior over the data:

$$\pi_{k+1}(x) = \mathbb{E}_{y\sim\mu}[p_k(x|y)].$$

10.2 The EM Covariance Update

The paper shows that the EM update for covariances is:

$$\tilde{\Sigma}_{k+1} = \tilde{\Sigma}_k + M_k(\Sigma - \tilde{\Sigma}_k)M_k, \qquad M_k = \tilde{\Sigma}_k(\tilde{\Sigma}_k + I)^{-1}.$$

Key Idea

Compare with the SCSI update:

$$\Sigma_{k+1} = \Sigma_k + B_k(\Sigma - \Sigma_k)B_k, \qquad B_k = \big(\Sigma_k(\Sigma_k + I)^{-1}\big)^{(1+\epsilon)/2}.$$

Setting $\epsilon = 1$ gives $B_k = (\Sigma_k(\Sigma_k + I)^{-1})^1 = M_k$. So:

$$\boxed{\textbf{EM is exactly the SCSI iteration with } \epsilon = 1.}$$

Since the optimal ϵ is 0, SCSI with ODE transport is *provably faster* than EM in this setting!

10.3 Non-Asymptotic Rates

Beyond the asymptotic (per-iteration) rate, the paper notes:

- EM enjoys a non-asymptotic rate of $O(1/k)$ due to its mirror-descent structure.
- Empirically, the ODE scheme achieves $O(1/k^2)$, reminiscent of Nesterov-type acceleration in convex optimization.

This is what Figure 2 in the paper illustrates: on a log-log plot, EM error decays with slope -1 (i.e., $1/k$) while ODE error decays with slope -2 (i.e., $1/k^2$).

11 Verifying the Theoretical Assumptions

11.1 Wasserstein Lipschitz Stability (Assumption 5)

Assumption 5 requires: $\mathbb{E}\|\Phi_\pi(y) - \Phi_{\tilde{\pi}}(y)\|^2 \leq R\, W_2^2(\pi, \tilde{\pi})$ for $R < 1$.

For Gaussians $\mathcal{N}(0, A)$ and $\mathcal{N}(0, B)$ with the ODE ($\epsilon = 0$), the transport cost is:

$$\mathcal{T}(A; B) := \mathrm{Tr}\big((T_A - T_B)(A + I)(T_A - T_B)\big),$$

where $T_A = (A(A + I)^{-1})^{1/2}$ is the ODE solution map.

The Wasserstein-2 distance between $\mathcal{N}(0, A)$ and $\mathcal{N}(0, B)$ is:

$$W_2^2(\mathcal{N}(0, A), \mathcal{N}(0, B)) = \mathrm{Tr}(A + B - 2(A^{1/2} B A^{1/2})^{1/2}).$$

Intuition

The paper shows numerically (Figure 3) that $\mathcal{T}(A; B) < W_2^2(A, B)$ with overwhelming probability for macroscopic SNR. In the extreme low-SNR limit ($\|A\|, \|B\| \to 0$), one can show R approaches 1 from below, so convergence slows but doesn't fail.

11.2 Condition Number (for KL Guarantees)

The condition number χ from Section 4 can be bounded explicitly:

$$\chi \leq (1 + \lambda_{\min}(\Sigma)^{-1})^2 \approx \eta^{-2} \quad \text{for small } \eta.$$

This captures the ill-conditioning of the inverse problem: when η is small (low SNR), the condition number is large, and more iterations are needed.

12 Summary

	SCSI (ODE, $\epsilon = 0$)	**EM ($\equiv$ SCSI with $\epsilon = 1$)**
Contraction factor	$\dfrac{1}{1+\eta}$	$1 - \left(\dfrac{\eta}{1+\eta}\right)^2$
Low-SNR rate	$(1-\eta)^k$	$(1-\eta^2)^k$
Non-asymptotic	$O(1/k^2)$ (empirical)	$O(1/k)$

The Gaussian AWGN case study shows that SCSI with ODE transport is the *optimal* choice in this family of methods, converging faster than both the SDE variant and the classical EM algorithm, especially in the challenging low-SNR regime.